Tables

Table 1. ANVISA corpus and regulatory use cases.

Regulation	Regulatory area	Example use case
RDC 945/2024 and RDC 9/2015	Clinical trials	DDCM procedures, DSUR timing, SUSAR handling, and trial-safety conduct.
RDC 406/2020	Pharmacovigilance	Minimum notification elements and follow-up when information is incomplete.
RDC 55/2010	Biological products	Registration rules and exceptions for foreign-manufactured biological products.
RDC 751/2022	Medical devices	Risk classification, labelling, and instructions for use.
IN 338/2024	Reliance and optimized review	Equivalent foreign regulatory authorities and reliance-related conditions.

Table 2. Development decisions made after observed failures.

Observed problem	Evidence from testing	Change made	Reason for regulatory use
Source text was noisy or poorly structured.	Early app tests and corpus review showed formatting artifacts and document-label problems.	Cleaned source text and verified document identity before indexing.	A regulatory answer is only useful if the cited source is identifiable.
Arbitrary chunks did not match legal structure.	Correct documents were sometimes nearby but answers missed article-level conditions.	Rebuilt the index with article-aware chunks and article metadata.	Users need checkable article citations, not only semantically similar passages.
Dense retrieval alone was weak for exact legal terms.	Terms such as DDCM, RDC numbers, article numbers, and IN 338/2024 required exact matching.	Added BM25 retrieval, reciprocal-rank fusion, and reranking.	Regulatory questions often depend on exact terms and article labels.
English questions over Portuguese documents lost completeness.	Initial English completeness was lower than Portuguese completeness.	Added conservative Portuguese retrieval anchors while preserving answer language.	International users may ask in English while the source corpus remains Portuguese.
The system answered from related but insufficient context.	External stress testing showed refusal correctness of 0.700.	Tightened the refusal rule and tested a separate refusal-focused external set.	A related article is not enough when the specific requested fact is absent.

Table 3. Representative answer outcomes.

Outcome	Example question	What the result showed
Correct direct answer	Which minimum information makes a notification valid and reportable?	The system retrieved RDC 406/2020, Art. 32 and gave the main required elements.
Correct false-premise handling	Is a notification valid without patient identification if the suspected medicine is named?	The system rejected the premise and cited RDC 406/2020, Art. 32.
Correct refusal	What is the current MedDRA version required today?	The system refused because the current required version was outside the indexed corpus.
Incomplete answer	What information must appear for non-printed instructions for use?	The system retrieved RDC 751/2022, Art. 51 but omitted that instructions must remain available while the product is on the market.
False refusal	Under RDC 945/2024, until when must DSURs be sent to Anvisa and what is the annual reference date?	One English external-test answer refused although the indexed corpus contained the answer.
Refusal failure before calibration	What is the current electronic form model for notifying SUSARs to Anvisa?	The earlier system answered from a related notification rule although the exact current form model was absent.